# Nishal Stanislaus Silva, *PhD*

Machine Learning Engineer | Anomaly Detection, Risk Modeling & Applied ML | PhD

*Canadian Permanent Resident, Eligible to work in Canada without sponsorship*

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## Professional Summary

Machine learning engineer with direct experience building financial forensics and anomaly-detection pipelines: at Forestpin, designed and deployed ML and statistical models for compliance monitoring and risk/anomaly detection on large structured enterprise datasets, with SQL + Python backend scoring services running in production for analyst triage. This is paired with a PhD-trained discipline in rigorous model evaluation – frozen-protocol benchmark design, ablation studies, and quantified accuracy/latency/CPU trade-offs – the same methodology that underpins validating a fraud or risk model before deployment. Broader ML engineering background spans real-time inference systems, embedded/edge deployment, and production ETL, most recently owning a full pipeline from acquisition through deployment and monitoring on an EU Horizon-funded project. Canadian Permanent Resident based in Toronto, available immediately.

## Core Competencies

**ML Engineering & Anomaly Detection:** statistical and ML anomaly/risk-scoring pipelines, compliance monitoring on structured enterprise data, SQL + Python production scoring services, feature engineering, model deployment and monitoring, ETL pipeline design.

**Research & Evaluation Rigor:** benchmark and ablation design, model evaluation methodology, quantified accuracy/latency/resource trade-off analysis, reproducible experiment protocols, technical writing and stakeholder translation.

## Technical Stack

Python (TensorFlow, Keras, PyTorch, scikit-learn, Pandas, NumPy, SciPy), C++17, SQL, AWS (EC2, S3, ECS, MWAA, RDS), Docker, Airflow, CI/CD, Git, ETL pipeline design.

## Education

**PhD, Information Engineering and Computer Science** – University of Trento, Italy (defended Jan 2025)
**MSc, Telecommunication and Electronic Engineering** – Sheffield Hallam University, UK (2018)
**BEng (Hons), Electronic Engineering** – Sheffield Hallam University, UK (2014)

## Portfolio

**Nebula** – C++17 audio-feature-extraction library with per-feature ARM/Raspberry Pi latency benchmarks (github.com/ns2max/neb

## Experience

**Independent Researcher** | *Self-directed, Toronto, Canada* *Jan 2026 - Present*

- Continued applied ML and systems work; two papers accepted for publication (JAES, IEEE IS2).
- Maintain Nebula, an open-source C++17 audio-feature-extraction library with per-feature ARM/Raspberry Pi latency benchmarks.
- Used AI-assisted development workflows, including Claude Code, for tooling and pipeline development.

**Postdoctoral Researcher** | *University of Trento, Trento, Italy* *Jan 2025 - Dec 2025*

- Owned the full ML pipeline for streaming audio and IMU/gesture data – acquisition, DSP/feature engineering, training and evaluation, edge deployment, and monitoring – on the EU Horizon EIC Pathfinder project MUSMET.
- Designed and deployed a real-time detector achieving 14 ms inference on Raspberry Pi 4 at F1 0.76 and 31.4% CPU utilization, 74x faster than the 1038 ms DTW baseline it replaced (published, JAES 2026).
- Built frozen-protocol benchmark suites and ablation studies quantifying accuracy/latency/CPU trade-offs, with a written record of negative results – methodology directly transferable to validating risk and fraud models before deployment.

**Visiting Researcher** | *McGill University (IDMIL/CIRMMT), Montreal, Canada* *May 2024 - Aug 2024*

- Built a self-powered smart electric guitar, fusing IMU and audio signals via a hierarchical finite-state-machine classifier, achieving F1 0.78 on a 500-event corpus (published, IEEE IS2 2025).

**Visiting Researcher** | *University of Visual and Performing Arts, Colombo, Sri Lanka* *Jan 2024 - Mar 2024*

- Conducted a human-centred evaluation study of musicians' perception of smart instrument pattern-detection systems (published, IEEE I3DA 2025).

**Technical Consultant** | *Forestpin (Pvt) Ltd, Colombo, Sri Lanka* *Aug 2020 - Feb 2021*

- Designed and deployed ML and statistical pipelines for financial forensics, compliance monitoring, and risk/anomaly detection on large structured enterprise datasets.
- Built SQL + Python backend anomaly-scoring services used in production for analyst triage.
- Translated compliance-stakeholder requirements into technical specifications for model development.

**Research Engineer** | *MAS Holdings (Pvt) Ltd, Colombo, Sri Lanka* *Jan 2015 - Jul 2020*

- Delivered end-to-end computer-vision and IoT systems for manufacturing at scale over 5.5 years, improving operational efficiency by up to 300%.
- Ran real-time IoT ETL pipelines across multiple sites with C++/Python inference on embedded hardware; introduced CI/CD and code review practices and mentored engineers.
- Automated multilingual care-label quality-control inspection (OpenCV, conveyor and PLC reject integration), cutting inspection time by 99.5%.

## Publications

Silva, N. & Turchet, L. (2026). Real-Time Audio Pattern Detection for Smart Musical Instruments. *Journal of the Audio Engineering Society*, 74(3). Silva, N. & Turchet, L. (2026, in press). Smart Drums. *Journal of the Audio Engineering Society*.

## Highlights

MIDI Innovation Awards 2023 Finalist. GMOA recognition (Government Medical Officers' Association, Sri Lanka, 2020) for a COVID-19 remote vitals-monitoring deployment.